Chapter 4.6 - Pipelined Datapath & Control

Comprehensive Revision Flashcards • 87 Key Concepts & Practice Questions

#1	**Q1: What is the first stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The first stage is Instruction Fetch, abbreviated as IF.

#2	**Q2: What is the second stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The second stage is Instruction Decode and register file read, abbreviated as ID.

#3	**Q3: What is the third stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The third stage is Execution or address calculation, abbreviated as EX.

#4	**Q4: What is the fourth stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The fourth stage is Data memory access, abbreviated as MEM.

#5	**Q5: What is the fifth stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The fifth stage is Write back, abbreviated as WB.

#6	**Q6: In a five-stage pipeline, up to how many instructions can be in execution during a single clock cycle?**
	Answer: Up to five instructions can be in execution during any single clock cycle.

#7	**Q7: What is the general direction of flow for instructions and data through the pipeline stages?**
	Answer: Instructions and data generally move from left to right through the five stages.

#8	**Q8: What is the first exception to the left-to-right flow of data in the pipelined datapath?**
	Answer: The write-back stage, which sends the result back to the register file in the middle of the datapath.

#9	**Q9: What is the second exception to the left-to-right flow of data in the pipelined datapath?**
	Answer: The selection of the next PC value, which chooses between the incremented PC and a branch address from the MEM stage.

#10	**Q10: The right-to-left data flow of the write-back stage can lead to what type of hazard?**
	Answer: This flow can lead to data hazards.

#11	**Q11: The right-to-left data flow for selecting the next PC value can lead to what type of hazard?**
	Answer: This flow can lead to control hazards.

#12	**Q12: To share a single datapath among multiple instructions in a pipeline, what hardware components must be added between stages?**
	Answer: Pipeline registers must be added to hold data between stages.

#13	**Q13: What is the naming convention for pipeline registers?**
	Answer: They are named for the two stages they separate (e.g., the register between IF and ID is named IF/ID).

#14	**Q14: Why is there no pipeline register at the end of the write-back (WB) stage?**
	Answer: It is redundant because all instructions must update some processor state (register file, memory, or PC) which already holds the result.

#15	**Q15: Which component of the processor's visible architectural state can be thought of as a pipeline register that feeds the IF stage?**
	Answer: The Program Counter (PC) can be thought of as a pipeline register feeding the IF stage.

#16	**Q16: What two pieces of information are placed into the IF/ID pipeline register during the Instruction Fetch stage?**
	Answer: The fetched instruction and the incremented PC address are placed in the IF/ID register.

#17	**Q17: During the ID stage of a load instruction, what values are stored in the ID/EX pipeline register?**
	Answer: The two read register values, the sign-extended immediate value, and the PC address are stored in the ID/EX register.

#18	**Q18: What calculation is performed by the ALU during the EX stage of a load (`ld`) instruction?**
	Answer: The ALU adds the contents of a register and the sign-extended immediate to calculate the effective memory address.

#19	**Q19: During the MEM stage of a load (\`ld\`) instruction, data is read from memory and placed into which pipeline register?**
	Answer: The data read from memory is placed into the MEM/WB pipeline register.

#20	**Q20: What is the final action performed during the WB stage of a load (\`ld\`) instruction?**
	Answer: Data is read from the MEM/WB pipeline register and written into the register file.

#21	**Q21: For a store instruction, what is passed from the ID/EX pipeline register to the EX/MEM pipeline register to be used in the MEM stage?**
	Answer: The register data to be stored is passed into the EX/MEM pipeline register.

#22	**Q22: What is the primary action during the MEM stage of a store (\`sd\`) instruction?**
	Answer: The data from the EX/MEM pipeline register is written into data memory.

#23	**Q23: What happens during the write-back (WB) stage for a store (\`sd\`) instruction?**
	Answer: Nothing happens in the write-back stage for a store instruction.

#24	**Q24: To prevent a structural hazard, each logical component of the datapath (e.g., ALU, data memory) can be used within how many pipeline stages?**
	Answer: Each logical component can be used only within a single pipeline stage.

#25	**Q25: What was the design bug in the initial pipelined datapath concerning the load (\`ld\`) instruction?**
	Answer: The write register number was not passed down the pipeline, so the wrong instruction would supply it during the WB stage.

#26	**Q26: How was the design bug for the load (\`ld\`) instruction fixed?**
	Answer: The destination register number is passed from the ID stage through the ID/EX, EX/MEM, and MEM/WB pipeline registers for use in the WB stage.

#27	**Q27: What is the key difference between a multiple-clock-cycle pipeline diagram and a single-clock-cycle pipeline diagram?**
	Answer: A multiple-clock-cycle diagram shows instruction execution over time, while a single-clock-cycle diagram shows the state of the entire datapath at one specific clock cycle.

#28	**Q28: A single-clock-cycle diagram represents a _____ slice of a multiple-clock-cycle diagram.**
	Answer: vertical

#29	**Q29: Why does allowing some instructions to take fewer cycles not necessarily improve pipeline throughput?**
	Answer: Throughput is determined by the clock cycle, not the latency (number of stages) of individual instructions.

#30	**Q30: Instead of making instructions take fewer cycles, what alternative approach could improve pipeline performance?**
	Answer: Making the pipeline longer so instructions take more, but shorter, clock cycles could improve performance.

#31	**Q31: Control signals for the EX, MEM, and WB stages are created during which pipeline stage?**
	Answer: The control signals are created during the instruction decode (ID) stage.

#32	**Q32: How are control signals transported to the correct pipeline stage for execution?**
	Answer: They are passed down the pipeline by extending the pipeline registers (ID/EX, EX/MEM, MEM/WB) to include control bits.

#33	**Q33: Which two control lines are set in the Execution/address calculation (EX) stage?**
	Answer: The ALUOp and ALUSrc control lines are set in the EX stage.

#34	**Q34: Which three control lines are set in the Memory access (MEM) stage?**
	Answer: The Branch, MemRead, and MemWrite control lines are set in the MEM stage.

#35	**Q35: Which two control lines are set in the Write-back (WB) stage?**
	Answer: The MemtoReg and RegWrite control lines are set in the WB stage.

#36	**Q36: The control signals created in the ID stage are first placed into which pipeline register?**
	Answer: They are placed into the ID/EX pipeline register.

#37	**Q37: Why is an instruction passed through a pipeline stage even if there is nothing for it to do in that stage?**
	Answer: This is done because later instructions are already progressing at the maximum rate, and there is no way to accelerate them.

#38	**Q38: What information must be preserved and passed down the pipeline for a load instruction to function correctly?**
	Answer: The destination register number must be preserved and passed through the pipeline registers to the WB stage.

#39	**Q39: In the stylized datapath diagrams, the register file is broken into two logical parts to represent its dual use in which two stages?**
	Answer: It is used for register reads during the ID stage and register writes during the WB stage.

#40	**Q40: What is the purpose of the ALUSrc control signal in the EX stage?**
	Answer: It selects whether the second ALU input is from a register (Read data 2) or the sign-extended immediate.

#41	**Q41: What is the purpose of the MemtoReg control signal in the WB stage?**
	Answer: It decides whether the value written to the register file comes from the ALU result or from data memory.

#42	**Q42: What are the five stages of a pipelined datapath in order?**
	Answer: 1. Instruction fetch (IF), 2. Instruction decode (ID), 3. Execution (EX), 4. Memory access (MEM), 5. Write back (WB).

#43	**Q43: What is the primary activity in the Instruction Fetch (IF) stage of a pipeline?**
	Answer: Reading an instruction from memory using the address in the Program Counter (PC).

#44	**Q44: What are the two main activities in the Instruction Decode (ID) stage of a pipeline?**
	Answer: Decoding the instruction and reading the contents of the specified registers from the register file.

#45	**Q45: What is the primary activity in the Execution (EX) stage of a pipeline?**
	Answer: Performing an operation in the ALU, such as an arithmetic calculation or an address calculation.

#46	**Q46: What is the primary activity in the Data Memory Access (MEM) stage of a pipeline?**
	Answer: Reading from or writing to the data memory.

#47	**Q47: What is the primary activity in the Write Back (WB) stage of a pipeline?**
	Answer: Writing the result of the instruction (from the ALU or memory) back into the register file.

#48	**Q48: In a five-stage pipeline, how many instructions can be in execution during a single clock cycle?**
	Answer: Up to five instructions.

#49	**Q49: What is the general direction of data and instruction flow through the pipelined datapath stages?**
	Answer: From left to right, from the IF stage to the WB stage.

#50	**Q50: What is the first exception to the left-to-right data flow in a pipelined datapath?**
	Answer: The write-back stage, which sends the result back to the register file located in the middle of the datapath.

#51	**Q51: What is the second exception to the left-to-right data flow in a pipelined datapath?**
	Answer: The selection of the next PC value, which can choose a branch address from the MEM stage.

#52	**Q52: The right-to-left data flow from the write-back stage can lead to what type of hazard?**
	Answer: Data hazards.

#53	**Q53: The right-to-left data flow for PC selection (branching) can lead to what type of hazard?**
	Answer: Control hazards.

#54	**Q54: What hardware components are added between the stages of a datapath to implement pipelining?**
	Answer: Pipeline registers.

#55	**Q55: The pipeline register located between the Instruction Fetch and Instruction Decode stages is named _____.**
	Answer: IF/ID.

#56	**Q56: What is the purpose of pipeline registers in a pipelined datapath?**
	Answer: To hold the data and control information for an instruction as it moves from one stage to the next, allowing stages to be shared by different instructions simultaneously.

#57	**Q57: Which component of the architectural state can be thought of as a pipeline register that feeds the IF stage?**
	Answer: The Program Counter (PC).

#58	**Q58: In the IF stage of a `ld` instruction, what happens to the PC value?**
	Answer: It is incremented by 4 and written back into the PC for the next cycle, and a copy is saved in the IF/ID pipeline register.

#59	**Q59: In the ID stage of a `ld` instruction, what values are stored in the ID/EX pipeline register?**
	Answer: The values from two read registers, the sign-extended immediate value, and the PC address.

#60	**Q60: What calculation is performed in the EX stage of a `ld` instruction?**
	Answer: The ALU adds the contents of a register to the sign-extended immediate value to calculate the memory address.

#61	**Q61: What happens in the MEM stage of a `ld` instruction?**
	Answer: The data memory is read using the address from the EX/MEM pipeline register, and the loaded data is placed in the MEM/WB pipeline register.

#62	**Q62: In the WB stage of a `ld` instruction, data is read from the _____ pipeline register and written into the register file.**
	Answer: MEM/WB

#63	**Q63: How does a `store` instruction make the data to be stored available during the MEM stage?**
	Answer: The data is read from the register file in the ID stage and passed through the ID/EX and EX/MEM pipeline registers.

#64	**Q64: What happens in the Write-Back (WB) stage for a `store` instruction?**
	Answer: Nothing happens in the WB stage for a store instruction, as the result was already written to memory in the MEM stage.

#65	**Q65: To avoid structural hazards, each logical component of the datapath, such as the ALU, can be used only within a single ____.**
	Answer: pipeline stage

#66	**Q66: What was the original design bug in handling the `ld` instruction's write-back stage?**
	Answer: The write register number was being supplied by the instruction in the IF/ID pipeline register, not the `ld` instruction that was actually in the WB stage.

#67	**Q67: How was the `ld` instruction bug fixed in the pipelined datapath design?**
	Answer: The destination register number is passed from the ID stage through the ID/EX, EX/MEM, and MEM/WB pipeline registers to be used in the WB stage.

#68	**Q68: What type of pipeline diagram shows the complete execution of multiple instructions over time, with clock cycles on one axis and instructions on the other?**
	Answer: A multiple-clock-cycle pipeline diagram.

#69	**Q69: What type of pipeline diagram shows the state of the entire datapath during a single clock cycle?**
	Answer: A single-clock-cycle pipeline diagram.

#70	**Q70: A single-clock-cycle diagram represents a vertical slice through what other type of diagram?**
	Answer: A multiple-clock-cycle diagram.

#71	**Q71: How are control signals for the EX, MEM, and WB stages passed down the pipeline?**
	Answer: They are generated in the ID stage and then stored in the control portions of the pipeline registers (ID/EX, EX/MEM, MEM/WB).

#72	**Q72: Which two control signals are set during the Execution/address calculation (EX) stage?**
	Answer: ALUOp and ALUSrc.

#73	**Q73: Which three control signals are set during the Memory access (MEM) stage?**
	Answer: Branch, MemRead, and MemWrite.

#74	**Q74: Which two control signals are set during the Write-back (WB) stage?**
	Answer: MemtoReg and RegWrite.

#75	**Q75: Why do the IF and ID stages not require special control signals to be passed to them?**
	Answer: Their primary operations (reading instruction memory, writing the PC, reading registers) are always asserted or are determined by fixed fields in the instruction format.

#76	**Q76: In the corrected datapath, the write register number for the WB stage comes from which pipeline register?**
	Answer: The MEM/WB pipeline register.

#77	**Q77: What information must be stored in the IF/ID pipeline register for a `beq` instruction to potentially work later on?**
	Answer: The incremented PC address.

#78	**Q78: During the EX stage of a `store` instruction, what two values are placed into the EX/MEM pipeline register?**
	Answer: The calculated effective address and the second register value (the data to be stored).

#79	**Q79: According to the 'Check Yourself' section, is it true that allowing some instructions to take fewer stages always increases pipeline performance?**
	Answer: No, because pipeline throughput is determined by the clock cycle time, not the latency of individual instructions.

#80	**Q80: The number of pipeline stages per instruction primarily affects an instruction's _____, not the pipeline's throughput.**
	Answer: latency

#81	**Q81: According to the 'Check Yourself' section, what is a potential benefit of making a pipeline longer (more stages)?**
	Answer: A longer pipeline can allow for a shorter clock cycle time, which could improve overall performance (throughput).

#82	**Q82: The IF/ID pipeline register must be wide enough to hold both the 32-bit instruction and the incremented ____ address.**
	Answer: 64-bit PC

#83	**Q83: The control logic generates control signals for the EX, MEM, and WB stages during which pipeline stage?**
	Answer: The Instruction Decode (ID) stage.

#84	**Q84: The control line _____ determines whether the ALU gets its second operand from the register file or the sign-extended immediate field.**
	Answer: ALUSrc

#85	**Q85: The control line _____ determines whether the value written to the register file comes from the ALU result or from data memory.**
	Answer: MemtoReg

#86	**Q86: The _____ control signal must be asserted to enable writing to the register file in the WB stage.**
	Answer: RegWrite

#87	**Q87: What two signals must both be asserted for the `PCSrc` multiplexor to select the branch target address?**
	Answer: The `Branch` signal (from control) and the `Zero` signal (from the ALU).